

Sahil Shaik

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Education

Georgia Institute Of Technology

Transferring Aug 2026

Bachelor of Science in Electrical Engineering

University Of Georgia

Aug 2025 – May 2026

Bachelor of Science in Electrical Engineering, GPA: 4.00

Relevant Courses: *Circuit Analysis, Fundamentals of Logic Design, Intro to Electrical and Computer Systems Eng.*

Work Experience

Data Science and Systems Informatics Lab @ UGA | Athens, GA

Aug 2025 – Current

Undergraduate Research Assistant – *Lab developing hardware for human-robot collaboration in industrial settings*

- Spearheaded electrical development of a robotic hand prototype designed to autonomously disassemble factory components, leading schematic and PCB design decisions for direct integration with the UR5e robotic arm
- Designed a custom servo controller PCB in Altium featuring a dual-stage power tree; integrated a high-current synchronous buck regulator to provide 6V motor power and a low-dropout (LDO) linear regulator supplying a stable 5V for the ATmega328P, enabling standalone operation from unregulated DC sources
- Incorporated overvoltage and reverse polarity protection into the PCB design to safeguard components against voltage fluctuations from the unregulated power supply and established serial communication with the UR5e system, enabling reliable command transmission during autonomous disassembly operations

System Technology Works LLC | Peachtree Corners, GA

May 2025 – Aug 2025

Electrical Engineering Intern – *Startup, developing entertainment robots for clients including Marvel and Paramount Pictures*

- Debugged manufactured PCBs using a systematic validation approach, isolating faults ranging from cold solder joints to shorted components by methodically verifying power rails, signal traces, and component integrity with a digital multimeter, consistently turning around fault diagnoses within 2-3 days
- Reviewed circuit schematics against manufactured PCB behavior to identify discrepancies between design intent and physical implementation, consistently identifying and documenting root causes that guided engineering team fixes and prevented recurring failures
- Developed interrupt-driven fault detection firmware for the motor control system of an entertainment robot on the NVIDIA Jetson Nano, configuring an ISR to monitor digital fault signals from the gate drivers and disabling PWM motor drive output within microseconds of a fault condition, preventing potential motor and gate driver damage during live operation

Projects and Activities

Wearable Haptic Navigation Glove

Jan 2026

- Self-initiated a haptic navigation glove project under faculty supervision with lab funding, designing a wearable prototype to provide directional proximity feedback to visually impaired users via 4 independently controlled vibration zones, accepted for demonstration at the UGA CURO Symposium
- Migrated from ESP32 to Teensy 4.0 (ARM Cortex-M7) for its higher ADC accuracy, resolving significant signal noise issues in the IR proximity sensing circuit and enabling reliable photodiode conduction current measurement across all 4 spatial zones
- Drove vibration motors using logic level N-channel enhancement MOSFETs switched by Teensy 4.0 GPIO PWM signals, enabling variable vibration intensity proportional to detected proximity without exceeding GPIO current limits

MicroMouse Club | Athens, GA

Aug 2025 – Current

National Competition Team – *Club developing an autonomous maze-solving robot for national competition*

- Jointly implemented competition board firmware, integrating motor control, PID, FloodFill navigation, IR sensor data acquisition, and BNO085 IMU and HC-06 Bluetooth modules over I2C and UART, producing a competition-ready autonomous maze-solving robot on the STM32 (ARM-Cortex M4F) target hardware
- Identified an ADC channel conflict arising from multiple sensors competing across a single ADC peripheral, implementing a DMA circular scan conversion to continuously cycle through channels and write results directly to memory for real-time firmware access without CPU intervention

Skills

Embedded Systems: STM32, Teensy 4.0, Arduino, ARM Cortex, NVIDIA Jetson Nano, I2C, UART, SPI, ADC, PWM, DMA, C, C++, Python, Verilog

Hardware: PCB Design, Schematic Design, MOSFETs, Oscilloscopes, Digital Multimeters, Logic Analyzers, Soldering

Software: Altium Designer, MATLAB, Simulink, EAGLE, SolidWorks